MIL-S-21521A (Wep)
24 August 1964
Superseding
MIL-S-21521 (Wep)
1 December 1960

MILITARY SPECIFICATION

SPRINGS, LIQUID

1. SCOPE

1.1 Scope. This specification states the requirements for three types of liquid springs for use in the Shock Absorber, Missile Launcher, Mark 2 Mod 0.

1.2 Classification. Liquid springs shall be of the following types:

Horizontal Type
Vertical Type U
Vertical Type L

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Sheet, Stock and Cut Shapes

FSC 144?

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PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof--for Packaging and Sealing

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-21568	Silicone Fluid, Dimethyl Polysiloxane

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

BuWeps LD 419651	Liquid Spring Assembly Vertical, Type U
BuWeps LD 419652	Liquid Spring Assembly Vertical, Type L
BuWeps LD 419653	Liquid Spring Assembly Horizontal

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2.1 (Cont)

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

NAVWEPS OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVWEPS QAP 9014	Shock Absorber, Missile Launcher, Mark 2 Mod 0
NAVORDINST 4355.1	Equipment for Use by Government Inspectors in Final Inspection of Naval Ordnance Items
BUWEPSINST 4355.5	Bureau of Naval Weapons Calibration Program
Standards Laboratory Information Manual	

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Description. A liquid spring consists of a cylinder, pistons, piston rod, and a charge of compressible fluid. The compressed fluid precludes movement of the pistons up to a predetermined load. When this load is exceeded, the liquid spring responds resiliently with the movement of the pistons against the compressible fluid, producing further compression which tends to return the pistons to the median position. Damping is attained by the use of orifices in the pistons which limit the rate of transfer of the fluid between the chambers of the cylinder.

3.2 General Requirements

3.2.1 Preproduction Samples. Unless otherwise specified in the contract or order, preproduction samples of the liquid springs shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to 4.2.1).

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the liquid springs shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWeps LD 419651, 419652, and 419653.

3.2.3 Finishes, Coatings, Construction, and Dimensions. Finishes, coatings, construction, and dimensions for the liquid springs shall be in accordance with the applicable drawings, specifications, and publications listed in BuWeps LD 419651, 419652, and 419653.

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3.2.4 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in BuWeps LD 419651, 419652, and 419653.

3.2.5 Interchangeability. Individual assemblies and piece parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with like numbered parts.

3.2.6 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Request for Deviations. The procedures for requesting deviations from this specification or any specification, publication, material, or process specified herein shall be as set forth in NAVWEPS OP 400.

3.2.9 Conflicting Requirements. In event of conflict between this specification and any specification, publication, or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances. Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

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3.2.11 Hydraulic Fluid. The hydraulic fluid used in the liquid springs shall conform to Specification MIL-S-21568 having a viscosity of 50 centistokes at temperature of 77 degrees Fahrenheit or, when so specified for leakage testing, dimethyl polysiloxane silicone fluid having a viscosity of 2 centistokes at temperature of 77 degrees Fahrenheit.

3.2.12 Charging Pressure. The internal charging pressure of the liquid springs, using the fluid specified in 3.2.11, shall be 13,500 ± 100 psig for the horizontal liquid spring and 18,000 ± 100 psig for each type of vertical liquid spring when gaged at a temperature of +80 ± 3 degrees Fahrenheit.

3.3 Performance Requirements

3.3.1 Static Load Deflection. Each liquid spring when charged as specified in 3.2.12 and while at a temperature of +80 ± 3 degrees Fahrenheit shall demonstrate conformance to the requirements of tables I, II, and III as appropriate for each type of liquid spring. Loading shall be applied to the longitudinal axis of the spring at a rate not less than 15 seconds for each stroke.

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TABLE I - STATIC DEFLECTION VS LOAD
AT +80 ± 3 DEGREES FAHRENHEIT

Liquid Spring, Horizontal Type

Spring Classification	Stroke	Direction of Piston Motion	Piston Displacement From Median (Inches)	Load Minimum (Pounds)	Load Maximum (Pounds)
Horizontal Type	Compression	Away from median	0-1/8	7,800	9,200
			1-1/2	11,700	13,500
			3	16,800	19,000
	Compression-Return	Toward median	3	15,000	17,200
			1-1/2	10,500	12,100
			0-1/8	7,100	8,200
	Extension	Away from median	0-1/8	7,400	8,800
			1-1/2	10,900	12,700
			3	15,300	17,500
	Extension-Return	Toward median	3	13,500	15,900
			1-1/2	9,500	11,300
			0-1/8	6,300	8,100

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TABLE II - STATIC DEFLECTION VS LOAD
AT +80 ± 3 DEGREES FAHRENHEIT

Liquid Spring, Vertical, Type U

Spring Classification	Stroke	Direction of Piston Motion	Piston Displacement From Median (Inches)	Load Minimum (Pounds)	Load Maximum (Pounds)
Vertical Type U	Compression	Away from median	0-1/8	8,300	9,400
			2-1/4	13,100	14,900
			4-1/2	19,200	21,700
	Compression-Return	Toward median	4-1/2	17,600	20,300
			2-1/4	12,300	13,900
			0-1/8	7,400	8,800
	Extension	Away from median	0-1/8	6,400	7,300
			2-1/4	9,200	10,600
			4-1/2	12,600	14,500
	Extension-Return	Toward median	4-1/2	11,300	13,400
			2-1/4	8,000	9,600
			0-1/8	5,500	6,700

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TABLE III - STATIC DEFLECTION VS LOAD
AT +80 ± 3 DEGREES FAHRENHEIT

Liquid Spring, Vertical, Type L

Spring Classification	Stroke	Direction of Piston Motion	Piston Displacement From Median (Inches)	Load Minimum (Pounds)	Load Maximum (Pounds)
Vertical Type L	Compression	Away from median	0-1/8	6,600	7,800
			2-1/4	9,500	11,000
			4-1/2	12,900	15,300
	Compression-Return	Toward median	4-1/2	12,100	13,500
			2-1/4	8,700	9,900
			0-1/8	6,000	7,200
	Extension	Away from median	0-1/8	7,500	8,500
			2-1/4	11,100	12,700
			4-1/2	15,800	17,400
	Extension-Return	Toward median	4-1/2	14,500	16,200
			2-1/4	10,300	11,700
			0-1/8	6,500	7,700

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3.3.2 Leakage With No Deflection (Accelerated). (See 6.1) When charged as specified in 3.2.12 (± 600 psig) with silicone fluid having a viscosity of 2 centistokes, the liquid spring shall show no loss in pressure in excess of 375 psi when allowed to stand under no deflection for 3 days. During that period, the spring temperature shall be not less than +70 degrees nor more than +83 degrees Fahrenheit and shall be maintained at a constant temperature ±4 degrees Fahrenheit. Thereafter, when charged with silicone fluid having a viscosity of 50 centistokes, the liquid spring shall be capable of meeting all performance requirements of this specification. The pressure loss shall be calculated by the following formula:

$$P = P_1 - P_2 + C\,(t_2 - t_1)$$

Where:

P = Pressure loss - (psig)

P_1 = Initial pressure - (psig)

P_2 = Final pressure - (psig)

C = Temperature coefficient = 70

t_1 = Initial temperature of spring degrees Fahrenheit

t_2 = Final temperature of spring degrees Fahrenheit

3.3.3 Leakage With No Deflection (Normal). (See 6.1) When charged as specified in 3.2.12 with silicone fluid having a viscosity of 50 centistokes, the liquid spring when allowed to stand under no deflection for 90 days at temperatures not less than +70 degrees nor more than +100 degrees Fahrenheit shall show no less in pressure in excess of 750 psi when measured at a spring temperature of 83 ± 3 degrees Fahrenheit. Thereafter, the liquid spring may be recharged, if necessary, and shall be capable of meeting all performance requirements of this specification.

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3.3.4 Dynamic Force and Deflection. When charged as specified in 3.2.12, each liquid spring shall be mounted in a drop-test machine as illustrated in figure 1. While maintained at a temperature of +80 ± 3 degrees Fahrenheit, the liquid spring shall be accelerated axially, once in compression and once in extension, by impact of the swinging weight acting upon the carriage. Deflection shall not exceed the limits in table IV. Thereafter, the liquid spring shall show no damage and shall meet all performance requirements of this specification.

TABLE IV

DYNAMIC FORCE AND DEFLECTION LIMITS

	Extension		Compression	
Liquid Spring	Maximum Force (Pounds)	Maximum Deflection (Inches)	Maximum Force (Pounds)	Maximum Deflection (Inches)
Horizontal Type*	24,500	3.4	24,000	3.5
Vertical Type U	16,000	4.7	22,000	3.5
Vertical Type L	24,000	3.6	16,000	4.8

*The force exerted at the piston-rod eye of the horizontal type spring during compression shall be not less than 19,000 pounds at 0.667-inch deflection from the median.

Characteristics of the drop-test machine used in this test shall be:

(a) Weights:

1. Swinging weight 3,224 ± 50 pounds
2. Suspension rods 750 ± 50 pounds
3. Carriage 550 ± 50 pounds

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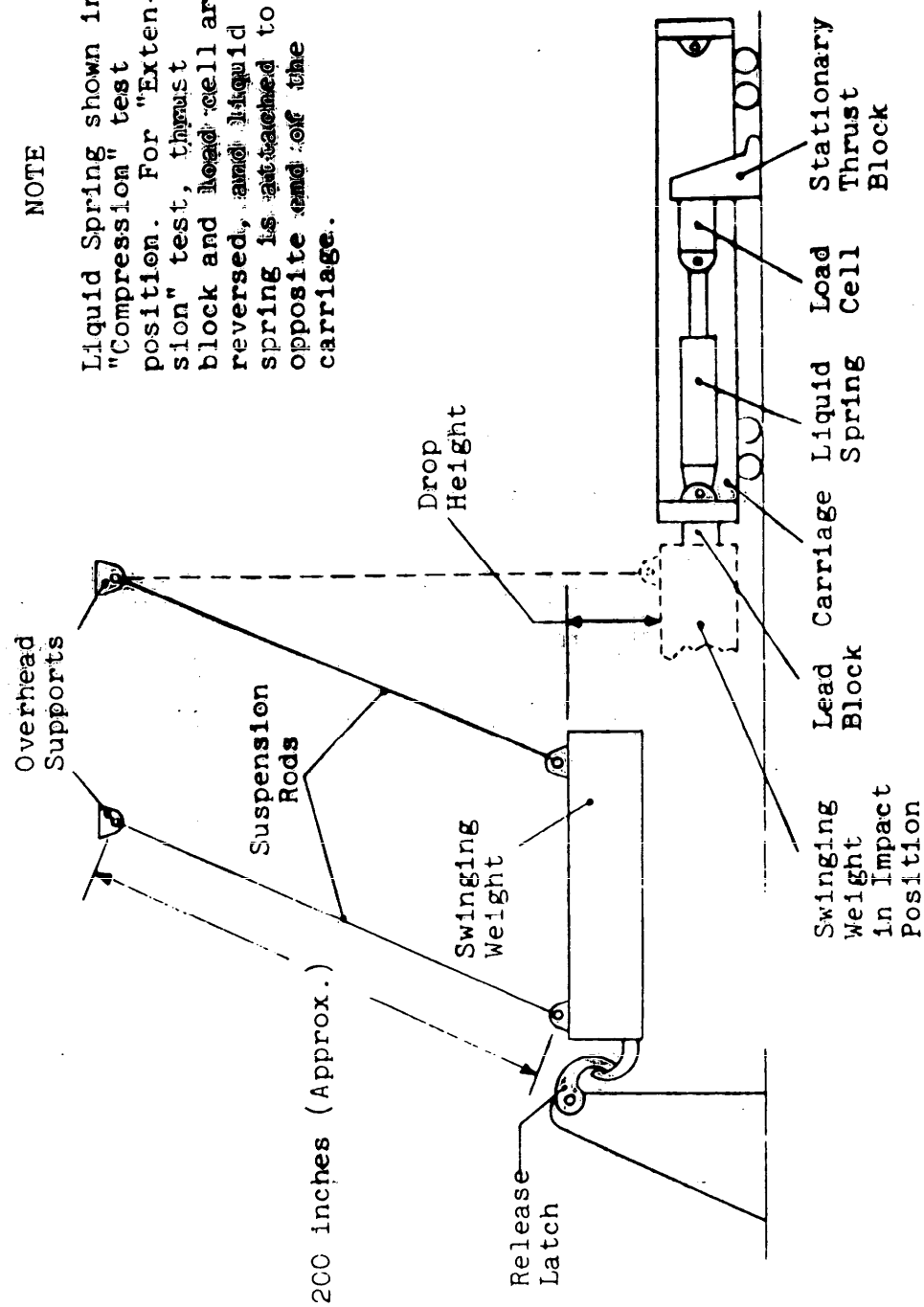


Figure 1. Liquid Spring Dynamic Force and Deflection Test Arrangement

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3.3.4 (Cont)

(b) Natural frequency of the test fixture shall be not less than 150 cps.

(c) Drop height of the swinging weight shall be 20.6 ± 0.5 inches.

(d) Maximum velocity of the carriage shall be attained in less than 6 milliseconds from standstill.

NOTE

The above conditions will cause the carriage to attain a maximum velocity of approximately 8-1/2 feet-per-second.

3.4 Environmental Conditions. For the environmental conditions specified in 3.4.1.1, 3.4.1.2, and 3.4.2 through 3.4.7, the liquid springs shall be charged as specified in 3.2.12. The liquid springs may be recharged before beginning each of the tests specified in 4.4.2.

3.4.1 Vibration, Operating Condition

3.4.1.1 Vertical Liquid Springs. The vertical liquid springs (types U and L) shall withstand the type I environmental vibration as specified in Standard MIL-STD-167, except that the frequency range shall be 5 to 25 cycles per second. During vibration, the temperature of the spring shall be maintained between +70 and +100 degrees Fahrenheit. Thereafter, the spring shall show no damage and may be recharged, if necessary, and shall meet all performance requirements of this specification, except that a pressure loss four times greater than that specified in 3.3.2 or 3.3.3 is permissible. The spring shall be mounted vertically in the vibration apparatus with the cylinder body eye attached to the apparatus and a 5,300-pound weight suspended from the piston rod eye. Vibration shall be applied at the cylinder body eye parallel to the longitudinal axis of the spring.

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3.4.1.2 Horizontal Liquid Spring. The horizontal liquid spring shall withstand the type I environmental vibration as specified in Standard MIL-STD-167, except that the frequency range shall be 5 to 25 cycles per second. During vibration, the temperature of the spring shall be maintained between +70 and +100 degrees Fahrenheit. Thereafter, the spring shall show no damage and may be recharged, if necessary, and shall meet all performance requirements of this specification, except that a pressure loss four times greater than that specified in 3.3.2 or 3.3.3 is permissible The spring shall be mounted horizontally in the vibration apparatus with the body eye attached to the apparatus and the piston rod eye attached to a 5,700-pound suspended weight unrestricted to movement parallel to the longitudinal axis of the spring. Vibration shall be applied at the cylinder body eye parallel to the longitudinal axis of the spring.

3.4.2 Vibration, Shipping Condition. The liquid springs, packaged and packed in accordance with section 5 herein, shall withstand each of the following vibration conditions for 10 minutes and thereafter shall show no damage and shall meet all performance requirements of this specification.

VIBRATION CRITERIA (MAX) FOR PACKAGING

Type of Transportation	Vertical		Longitudinal		Lateral	
	cps	g	cps	g	cps	g
Truck	200	2.5	150	1.5	200	1.0
	10	1.0	10	1.0	10	1.0
Rail	60	2.0	60	1.0	30	1.0
	10	1.0	10	0.5	10	0.5
Aircraft	60-100	1.0	60-100	1.0	60	1.0
	100-200	0.5			100	0.5
Ship (surface)	10	0.5	Negligible		10	0.25
	30	1.5			30	0.75

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t = Initial temperature of spring

3.4.3 Lateral Shock, Operating Condition. The liquid springs shall withstand six shock applications resulting from impact of the spring (at centerline of cylinder body eye) against a stop. The liquid spring shall be suspended at the piston rod eye from a supporting fixture. The lower end of the cylinder shall be raised and allowed to drop freely from a height which shall produce a step velocity of 12 ± 1 feet per second at the moment of impact. Upon impact, the velocity shall decrease from maximum to zero within 10 milliseconds. After the sixth drop, the spring shall show no damage and thereafter shall meet all performance requirements of this specification.

3.4.4 Shock, Shipping Condition. The liquid springs, when packaged and packed in accordance with section 5 herein, shall withstand the following shock conditions and thereafter shall show no damage and shall meet all performance requirements of this specification.

SHOCK CRITERIA FOR PACKAGING
(Random loads including gravity)
Acceleration g (max)

Type of Transporation	Parallel to Direction of Transportation	Lateral	Vertical
Rail	20	2	5
Aircraft or Truck	5	1	3

3.4.5 Corrosive Atmosphere. The liquid springs shall withstand the salt spray test specified in Procedure I of Specification MIL-E-5272. Thereafter, the liquid springs shall show no corrosion detrimental to operation or damage and shall meet all performance requirements of this specification.

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3.4.6 Operation Under Casualty Temperature Condition. The liquid springs shall be subjected to 5 cycles of operation under a casualty temperature condition of +110 ± 5 degrees Fahrenheit. Thereafter, the liquid springs shall meet the requirements of 3.3.1. A cycle of operation shall be the displacement of the piston rod from the median position to extension and return to median; then to compression and return to the median position. Piston rod displacement and loading shall conform to tables V, VI and VII. Loading shall be in the longitudinal axis of the spring at a rate of not less than 1 minute for each complete cycle.

3.4.7 Operating Life. The liquid springs shall be subjected to and shall withstand 1,000 cycles of operation without maintenance. During the cycles of operation, the temperature of the spring shall be maintained between +70 and +100 degrees Fahrenheit; the duration of piston travel through one-quarter cycle shall be not less than 5 seconds; and the loss of fluid pressure shall be not greater than 2,100 psig. Thereafter, the spring may show signs of wear and damage and, after recharging to the specified charging pressure (3.2.12), shall meet the performance requirements specified in 3.3.1 and 3.3.2 (or 3.3.3). A cycle of operation shall be as follows:

Vertical Liquid Springs:	From median position to extension of 5-5/8 ± 1/8 inches, to median position, to compression of 5-5/8 ± 1/8 inches, and return to median.
Horizontal Liquid Springs:	From median position to extension of 3-5/8 ± 1/8 inches, to median position, to compression of 3-5/8 ± 1/8 inches, and return to median.

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TABLE V - STATIC DEFLECTION VS LOAD
AT 110 ± 5 DEGREES FAHRENHEIT

Liquid Spring, Horizontal Type

Spring Classification	Stroke	Direction of Piston Motion	Piston Displacement From Median (Inches)	Load Minimum (Pounds)	Load Maximum (Pounds)
Horizontal Type	Compression	Away from median	0-1/8	8,800	10,600
			1-1/2	12,900	15,000
			3	18,000	20,300
	Compression-Return	Toward median	3	16,200	18,900
			1-1/2	11,800	13,900
			0-1/8	8,200	9,900
	Extension	Away from median	0-1/8	7,900	10,200
			1-1/2	11,400	14,300
			3	15,800	18,800
	Extension-Return	Toward median	3	15,100	17,700
			1-1/2	10,600	12,900
			0-1/8	7,400	9,500

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TABLE VI - STATIC DEFLECTION VS LOAD AT 110 ± 5 DEGREES FAHRENHEIT

Liquid Spring, Vertical, Type U

Spring Classification	Stroke	Direction of Piston Motion	Piston Displacement From Median (Inches)	Load Minimum (Pounds)	Load Maximum (Pounds)
Vertical Type U	Compression	Away from median	0-1/8	8,600	11,000
			2-1/4	13,400	16,600
			4-1/2	19,700	23,400
	Compression-Return	Toward median	4-1/2	19,000	22,300
			2-1/4	13,000	15,800
			0-1/8	8,200	10,700
	Extension	Away from median	0-1/8	7,300	8,900
			2-1/4	10,200	12,300
			4-1/2	13,700	16,500
	Extension-Return	Toward median	4-1/2	12,500	15,100
			2-1/4	9,100	11,400
			0-1/8	6,400	8,400

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TABLE VII - STATIC DEFLECTION VS LOAD
AT 110 ± 5 DEGREES FAHRENHEIT

Liquid Spring, Vertical, Type L

Spring Classification	Stroke	Direction of Piston Motion	Piston Displacement From Median (Inches)	Load Minimum (Pounds)	Load Maximum (Pounds)
Vertical Type L	Compression	Away from median	0-1/8	7,100	9,600
			2-1/4	10,100	13,200
			4-1/2	13,500	17,200
	Compression-Return	Toward median	4-1/2	13,000	15,700
			2-1/4	9,800	11,900
			0-1/8	6,900	9,100
	Extension	Away from median	0-1/8	8,600	10,800
			2-1/4	12,200	14,700
			4-1/2	16,900	19,900
	Extension-Return	Toward median	4-1/2	15,900	19,200
			2-1/4	11,500	14,100
			0-1/8	7,800	10,200

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3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with this specification. Particular attention shall be paid to neatness and thoroughness of work, marking of parts and assemblies, plating, painting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Inspection. Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Government Inspection. The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in NAVWEPS OP 400. Production tests of final assemblies shall be on a 100-percent basis. No sampling is permitted for acceptance inspection purposes. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with NAVWEPS OP 400. Failed parts and components shall be submitted to the Material Review Board for disposition.

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4.2.1 Preproduction Samples. The preproduction samples specified in 3.2.1 shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract, or by the procuring activity, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order. Disposition of the test sample shall be at the discretion of the Material Review Board.

4.3 Classification of Tests. The inspection and testing of the liquid springs shall be classified as follows:

4.3.1 Production Tests. These tests are to be accomplished on each liquid spring being submitted for acceptance under the contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector or these tests may be conducted by the Government Inspector. These tests are detailed in NAVWEPS QAP 9014.

4.3.2 Preproduction Tests. Preproduction tests are those which are accomplished on samples selected as specified in 4.2.1, which are representative of the production of the liquid springs, to determine that the production meets the requirements of this specification. These are detailed in NAVWEPS QAP 9014. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction is approved by the procuring activity.

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4.3.3 Tests After Delivery. Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the liquid springs. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.4 Acceptance Testing. Tests shall be conducted to demonstrate compliance of the liquid springs with each requirement of section 3.3 and 3.4. Testing shall be in accordance with applicable sections of NAVWEPS QAP 9014.

4.4.1 Production Tests. The contractor or Government Inspector shall conduct production tests as specified in 4.3.1 to assure that the liquid springs are in compliance with the requirements of 3.3.1, 3.3.2 (or 3.3.3), and 3.3.4.

4.4.1.1 Test Conditions. Unless otherwise specified, the liquid springs shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, 80 ± 3 degrees Fahrenheit
Altitude:	14.7 ± 3 psia
Vibration:	1/4g ambient acceleration or less
Humidity:	Ambient

4.4.1.2 Test and Inspection Equipment and Facilities. The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required production tests.

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4.4.1.2.1 Calibration. Measuring equipment and instrumentation shall be calibrated as required by NAVORDINST 4355.1, BUWEPSINST 4355.5, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by the Bureau of Naval Weapons Representative, Pomona, California.

4.4.2 Preproduction Tests. Preproduction tests will be conducted as specified in 4.3.2 to assure compliance with the requirements of 3.4.1 through 3.4.7.

4.5 Rejection. Failure of the liquid springs to meet any of the requirements or tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be level A or C as specified (see 6.3).

5.1.1 Level A

5.1.1.1 Cleaning. The liquid spring shall be cleaned in accordance with Method C-1 of Specification MIL-P-116.

5.1.1.2 Drying. The liquid spring shall be dried in accordance with one or more of the methods given in Specification MIL-P-116, provided the method is not injurious to the equipment.

5.1.1.3 Preservative Application. None

5.1.1.4 Unit Packaging. The liquid spring shall be packaged in accordance with Method III of Specification MIL-P-116. Cover each end of the cylinder with a transparent polyethylene pouch, 6 mil minimum thickness, of such length that the open end of one pouch overlaps the other approximately 2 inches at the center. Tape the pouches together at center, using pressure sensitive tape conforming to Specification PPP-T-60.

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5.1.1.5 Cushioning. The package shall be cushioned by a top pad and a bottom pad built up of corrugated fiberboard conforming to Specification PPP-F-320, Type CF, domestic, DW grade 200. Each item shall be die cut to appropriately conform to the liquid spring contour to a depth of approximately 1-1/8 inches; the total thickness of the pad shall be about 1-1/2 inches.

5.1.2 Level C. Level C preservation and packaging shall be in accordance with 5.1.1.1, 5.1.1.2, 5.1.1.3, 5.1.1.4, and 5.1.1.5.

5.2 Packing. Packing shall be level A or level C as specified (see 6.3).

5.2.1 Level A. The package and cushioning shall be packed in a close-fitting container conforming to Specification PPP-B-636, Type I (CF), class 2, grade W5C, style RSC.

5.2.2 Level C. Same as 5.2.1 except that container may be class 1.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Leakage. Each liquid spring shall be tested to meet the requirements of either 3.3.2 or 3.3.3, but not both. The requirements of 3.3.3 shall ordinarily be met but when time limitations so dictate the alternate requirement of 3.3.2 will be acceptable.

6.2 Intended Use. The liquid spring is intended for use in the Shock Absorber, Missile Launcher, Mark 2 Mod 0, which is a component of the Launching System, Missile Mark 17 Mods 0, 1, and 2, and Mark 21 Mods 0 and 2.

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6.3 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Exceptions to the specification, applicable drawings, and other documents.

c. Selection of applicable levels of preservation, packaging, and packing.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-21521A(Wep) SPRINGS, LIQUID

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426

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